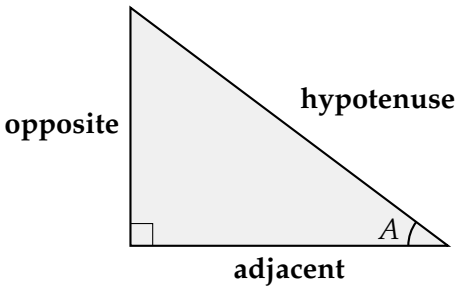
First Steps in Trigonometry

Sine, Cosine and Tangent — the secret ratios inside every right triangle

The big idea

Trigonometry ("triangle measuring") lets you find a length you *can't* measure — the height of a tree, a building, a kite — from one length and one angle you *can* measure.
It works because in a right triangle, **the angle decides the shape**, and the shape decides the **ratios** of the sides.

1 Name the sides — always from the angle you are looking at

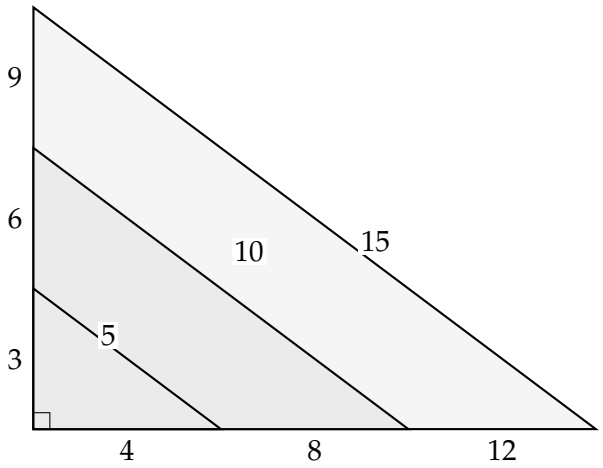


Stand at the angle A (the red arc) and look into the triangle:

- **Hypotenuse** — the longest side, **across from the right angle** ⌞. It never changes.
- **Opposite** — the side **across from** angle A. It doesn't touch A.
- **Adjacent** — the side **next to** angle A that is *not* the hypotenuse. ("Adjacent" means "next to.")

Watch out: opposite and adjacent **swap** if you stand at the other sharp angle. Always ask first: "Which angle am I looking from?"

2 Why ratios? A simple case with three triangles



These are **scale drawings** of one triangle (scale factors 1, 2, 3), so they all share the same angles. Look at $\frac{\text{opposite}}{\text{hypotenuse}}$ for the bottom-right angle:

$$\frac{3}{5} = 0.6 \qquad \frac{6}{10} = 0.6 \qquad \frac{9}{15} = 0.6$$

Same angle $\Rightarrow$ **same ratio**, no matter how big the triangle is. These ratios get names: sine, cosine and tangent.

3 The three definitions: SOH CAH TOA

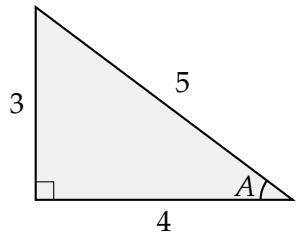
For an acute angle A in a right triangle

$$\sin A = \frac{\textbf{O}\text{pposite}}{\textbf{H}\text{ypotenuse}} \qquad \cos A = \frac{\textbf{A}\text{djacent}}{\textbf{H}\text{ypotenuse}} \qquad \tan A = \frac{\textbf{O}\text{pposite}}{\textbf{A}\text{djacent}}$$

SOH CAH TOA

Memory sentence: *Some Old Horses Can Always Hear Their Owners Approach.*

Example 1 · read the ratios



From angle A: opposite $= 3$, adjacent $= 4$, hypotenuse $= 5$.

$$\sin A = \frac{3}{5} = 0.6 \qquad \cos A = \frac{4}{5} = 0.8 \qquad \tan A = \frac{3}{4} = 0.75$$

Simple check: sine and cosine are always **less than 1**, because the hypotenuse is the longest side.

4 Use a table to find a missing side

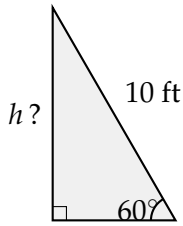
Angle	30°	37°	45°	53°	60°
sin	0.50	0.60	0.71	0.80	0.87
cos	0.87	0.80	0.71	0.60	0.50
tan	0.58	0.75	1.00	1.33	1.73

Values are rounded. (A scientific calculator gives them too — make sure it is in **degree** mode.)

Four steps

1. Mark the angle you know. **2.** Label the sides: hyp, opp, adj. **3.** Pick the ratio that uses the side you *know* and the side you *want*. **4.** Write the equation and solve it like a proportion.

Example 2 · how high does the ladder reach?



A 10-foot ladder leans on a wall at a 60° angle with the ground.
Know: hypotenuse $= 10$. Want: the height, which is **opposite** the 60° angle.
Opposite and hypotenuse $\Rightarrow$ **SOH**, sine:

$$\sin 60^\circ = \frac{h}{10} \;\Rightarrow\; 0.87 = \frac{h}{10} \;\Rightarrow\; h = 10 \times 0.87 = 8.7 \text{ ft.}$$

Example 3 · the unknown is on the bottom

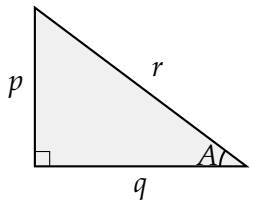
The opposite side of a 30° angle is 6. How long is the hypotenuse? $\sin 30° = \frac{6}{h} \Rightarrow 0.5 = \frac{6}{h} \Rightarrow h = \frac{6}{0.5} = 12$. **Check:** $\frac{6}{12} = 0.5$ ✓

5 Practice

A. Name the sides EASY

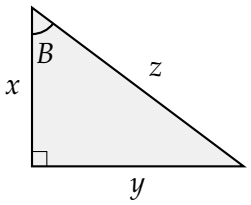
For the marked angle, write which side is the hypotenuse, the opposite and the adjacent.

1.



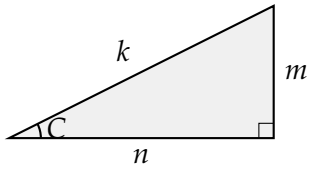
hyp ____ opp ____ adj ____

2.



hyp ____ opp ____ adj ____

3.

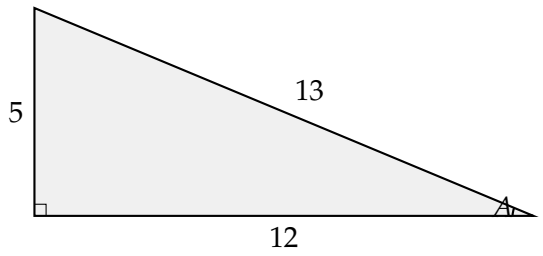


hyp ____ opp ____ adj ____

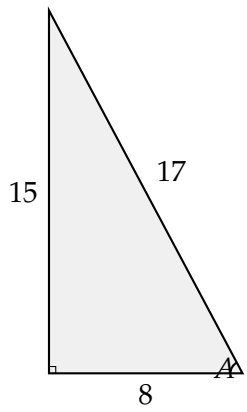
B. Write the ratios EASY MEDIUM

Write sin *A*, cos *A and* tan *A as fractions in lowest terms, then as decimals (round to the hundredths).*

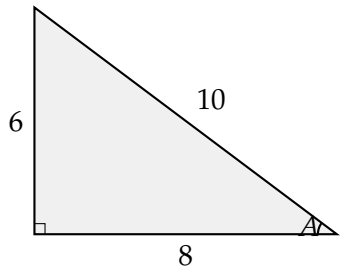
4.



5.



6.



7. MEDIUM In the triangle of problem 4, now stand at the **other** sharp angle B. Write $\sin B$, $\cos B$ and $\tan B$. What do you notice?

C. Same angle, same ratio MEDIUM

8. Triangle P has legs 5 and 12. Triangle Q has legs 10 and 24. Find tan of the angle across from the short leg in each. Are the two angles equal? Explain.

D. Which ratio? MEDIUM

Write sin, cos or tan. (Which two sides does the problem use?)

9. Know the hypotenuse, want the opposite.

10. Know the adjacent, want the opposite.

11. Know the hypotenuse, want the adjacent.

12. Know the opposite, want the hypotenuse.

E. Find the missing side MEDIUM HARD

Use the table in Part 4. Draw a picture, label hyp / opp / adj, then solve.

13. MEDIUM The hypotenuse is 20 and one angle is 30°. Find the side opposite the 30° angle.

14. MEDIUM A ramp is 10 m long and rises at an angle of 30°. How high does it go?

15. MEDIUM One leg is 8 and the angle next to it is 45°. Find the other leg (opposite the 45° angle).

16. MEDIUM The hypotenuse is 15 and one angle is 37°. Find the side adjacent to the 37° angle.

17. MEDIUM A tree casts a shadow 12 m long. From the tip of the shadow, the top of the tree is at an angle of 53°. How tall is the tree?

18. HARD The side opposite a 30° angle is 7. Find the hypotenuse.

19. HARD A kite string is 50 m long and makes a 37° angle with the ground. How high is the kite? How far away along the ground is it?

F. Think about it HARD

20. Why can $\sin A$ never be bigger than 1?

21. In a right triangle, $\sin A = \frac{3}{5}$ and the hypotenuse is 25. Find the opposite side. Then use $\cos A = \frac{4}{5}$ to find the adjacent side.

22. Why is $\tan 45^\circ$ exactly 1?

Remember: 1) Mark the angle. 2) Label hyp, opp, adj from that angle. 3) SOH CAH TOA. 4) Check: does the hypotenuse come out the longest?